

Act I · Develop

Principal Investigator

140 research points

Baby Mateo case · neural-crest spacing in the developing frontal bone

Student: **Jordan Rivera (sample student)**

Date: Example run, Act I

ABSTRACT

Write this LAST: an abstract is a short summary of the whole report (about 150 to 300 words) that answers what you did, why, how, what you found, and what it means. The draft below is built from your run; refine it in your own words.

Do neural crest cells space themselves non-randomly as they build the frontal bone? Where these cells settle helps build the frontal bone, so whether they arrange themselves on purpose is worth testing. To test this, we measured the nearest-neighbor spacing of cell nuclei in FN1-rich tissue, comparing it against FN1-blocked (control) and a known-ordered positive control, across 3 embryos. We found that the FN1-rich tissue was more spread out than the FN1-blocked (control) (mean relative spacing 0.0843 versus 0.0450; honest per-replicate test $p < 0.001$, $n=3$). We conclude that nuclei are non-randomly spaced in FN1-rich tissue compared to control, because the difference from the control is unlikely to be due to chance. We measured spacing, not force, so this shows the cells are organized, not that anything is pulling them: spatial organization is not the same thing as tension.

01 · QUESTION (ASK)

Do neural crest cells space themselves non-randomly as they build the frontal bone?

02 · HYPOTHESIS & PREDICTION

A graded FN1 “road” directs neural crest migration, producing ordered nuclear spacing.

PREDICTION FN1-rich tissue shows larger, more regular spacing than FN1-blocked control.

03 · TOOLS & BUDGET

Tools selected	Spatial layout
Evidence ceiling	Spatial layout
Team hired	none
Grant spent	\$0 of \$50,000

04 · DESIGN

Control	FN1-blocked tissue (negative control) + a known-ordered sample (positive control)
Replicates	3 embryos
Honesty label	2D-projected (labeled honestly)

05 · MEASUREMENTS (RUN)

FN1-rich tissue · mean spacing (rel.)	0.0843
FN1-blocked (control) · mean spacing (rel.)	0.0450
Replicates (real n)	3
Honest per-replicate test	$p < 0.001$ · significant

06 · ANALYSIS

Only the 3 replicates count as the real sample ($n=3$), and the honest per-replicate test already finds a clear difference ($p < 0.001$). Pooling treats all 599 cells across those replicates as if each were its own experiment, inflating the sample far past 3. That inflation is what makes a pooled result look more certain than the evidence supports, which is pseudoreplication, so the per-replicate test is the one to trust. Spacing here is a relative nearest-neighbor index in normalized image units; a larger number means cells sit farther apart.

07 · CONCLUSION · CLAIM · EVIDENCE · REASONING · LIMITATION

SUPPORTED

CLAIM Nuclei are non-randomly spaced in FN1-rich tissue compared to control.

EVIDENCE Nearest-neighbor spacing (relative, normalized units; larger = farther apart), FN1-rich tissue vs FN1-blocked (control), measured per replicate ($n=3$). Honest test $p < 0.001$; mean 0.0843 vs 0.0450. Control: FN1-blocked tissue (negative control) + a known-ordered sample (positive control).

REASONING Spacing is exactly what this claim is about, and your control isolates the FN1 road, so a spacing difference is real evidence about FN1. The claim says only what your most direct tool actually measured, so it is supported.

LIMITATION Spatial organization is not the same thing as tension. Even with more money, you can't make a tension claim here: the machine isn't in this lab, and your embryo is alive.

08 · WHAT I CORRECTED (THE 6 RS · 2 STEPS REVISED)

Ask

REVISED I started with "why do babies get clefts," but that is a why I cannot measure. I changed it to whether the crest cells space themselves non-randomly, which I can actually count.

NEXT Write the testable version of my question first, before I get attached to the big one.

Conclude

REVISED My first instinct was to say FN1 pulls the cells into place. I changed it to "the cells are non-randomly spaced," because I measured spacing, not force.

NEXT Match the claim to the tool I actually used, and name out loud the thing I did not measure.

BADGES EARNED

Did the Reading · 15

Sharp Question · 10

Prediction Made · 10

Tooled Up · 10

Fair Design · 15

Verified Measurement · 15

Honest Ceiling · 25

Self-Corrector · 20

Closed the Loop · 20

Generated locally from this run. No data left the browser. Develop · Differentiate · Derail ·
mendozabiomed.org